Great Wall

Coercion-resistant authentication

Yuri S. Villas Boas · author, Bitcoin BIP-450

This is not a foreign problem

344

documented physical attacks on Bitcoin holders

12

of them in Brazil

2

this year

Florianópolis, 2017. Jameson Lopp's public registry.

Our answer has been:

“Don’t tell anyone.”

What actually works

You cannot be forced to hand over

something you cannot describe.

Recognition, not recall. There is nothing to write down, and nothing to take.

How it looks to a user

Four palettes. Sixteen faces each. Reshuffled every time.

You pick the one you know.

Demo: `youtube.com/watch?v=9lJvw-8PZOA`

Two deployments, one protocol

Custodial

2^{16}

The bank counts your guesses and freezes the account.

Sixteen bits is plenty.

Self-custodial

2^{160}

Nobody counts guesses against your own encrypted seed.

Which is why it can't be faces.

Where the money is

License it to banks

$80.8M

serviceable market, 673 institutions

Brazil's central bank already imposes night-time Pix limits. It has conceded that **delay** is the countermeasure. Its own data shows the refund mechanism recovering ~9% of confirmed fraud.

Sell it to holders

$19.4M

serviceable market

The self-custody marketplace. Smaller, and it is where the credibility comes from.

The ask

$600K

for 20% at $3.0M post-money

17 months of runway — team, integration, first bank pilot

backup

Why me

- Author of **BIP-450** — a Bitcoin standard, peer-reviewed by the same process
- **37-page paper** formalising the attack, with a public DOI
- Contributor to the attack registry the model is calibrated on
- `great-wall-core` is **public and in beta** — the encoder is real code

Solo founder. Pre-incorporation — I incorporate on term sheet, in whichever jurisdiction the round wants.

“Doesn’t multisig solve this?”

No — it moves the problem.

A co-signer who can be reached is a second victim, not a second lock.

A co-signer who *cannot* be reached is a custodian, and you have given up self-custody to get there.

“What about a duress PIN?”

That is the failure mode, not the fix.

A partial disclosure the attacker cannot verify gives them a reason to keep going. Graceful degradation — a virtue everywhere else in engineering — is, under coercion, an incentive to escalate.

Blockstream ships a duress PIN. That is the strongest evidence I have that the problem is real.

Status, honestly

- Self-custodial encoder: **public beta**
- Custodial deployment: **designed, not built** — and it is the *smaller* engineering problem
- No entity yet; no revenue yet

The seed builds the custodial product and lands the first bank pilot.